Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 25)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

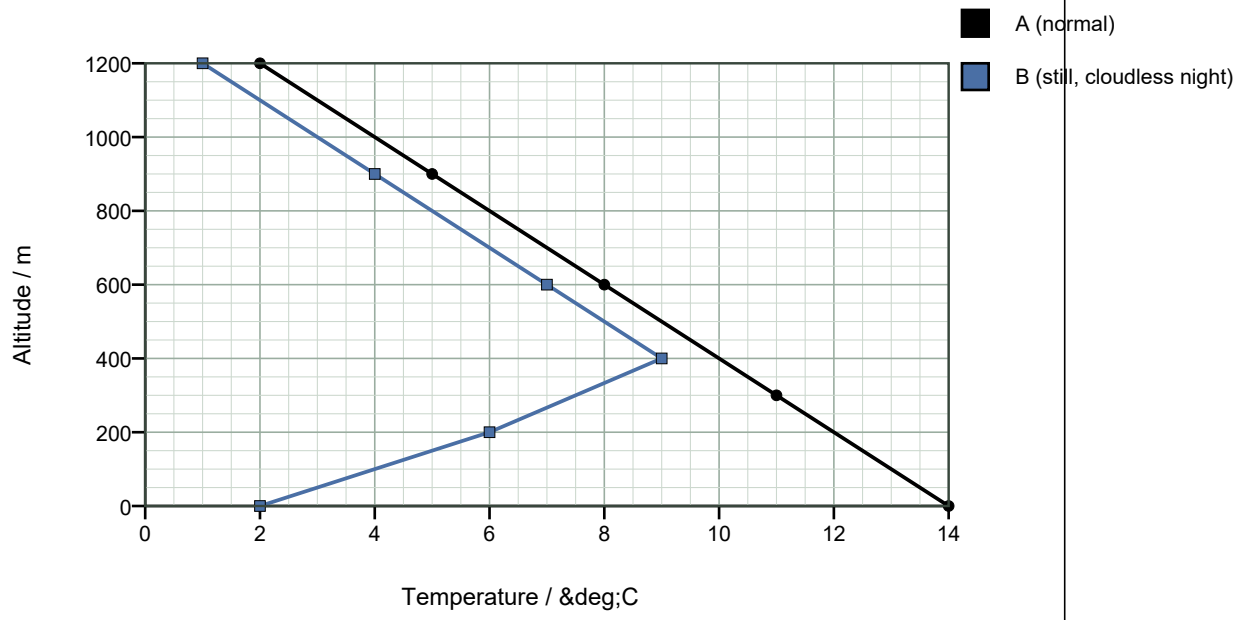
Question	Mark
1	
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TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows how air temperature changes with altitude above a city in a valley on two different nights. Line A shows normal conditions. Line B shows the conditions on a still, cloudless night.

Figure 1



0 1 . 1 Name the atmospheric condition shown by line B in Figure 1.

[1 mark]

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0 1 . 2 Use Figure 1 to explain why pollutants released at ground level would disperse on night A but not on night B.

[4 marks]

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0 1 . 3 Explain why the conditions on night B (still, cloudless) allowed an inversion to form.

[3 marks]

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0 1 . 4 Give **two** types of pollution event that are made more severe by temperature inversions.

[2 marks]

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2 .....

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0 2 A 1960s office block is being refurbished. Table 1 shows the measured heat loss through its fabric before the work.

Table 1

Part of the building	Heat loss / kW
Walls (uninsulated)	42
Roof	26
Single-glazed windows	31
Floor	9
Draughts and ventilation	22

0 2 . 1 Calculate the percentage of the total heat loss that passes through the windows.

[2 marks]

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Answer [ ] %

0 2 . 2 Explain how each of the following reduces heat loss.

[3 marks]

Cavity wall insulation .....

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Low-emissivity double glazing .....

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A heat exchanger on the ventilation system .....

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0 2 . 3 Explain why there is an optimum thickness of loft insulation rather than simply using as much as possible.

[2 marks]

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0 2 . 4 The refurbishment also includes a combined heat and power (CHP) unit. Explain why a CHP unit has a lower **electrical** efficiency than a large power station but a much higher **overall** efficiency.

[3 marks]

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0 2 . 5 Explain **five** other ways the refurbished building could be designed or managed to use less energy.

[5 marks]

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0 3 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☒

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 3 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

0 3 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

1 .....

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0 4 . 1 Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

0 4 . 2 Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 4 . 3 Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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0 5 . 1 Explain how hydrogen can be used to store energy from intermittent renewable resources, and describe **two** ways the stored energy can be released.

[4 marks]

0 5 . 2 Compare **three** methods of storing hydrogen, giving a disadvantage of each.

[3 marks]

0 5 . 3 Explain how a fuel cell differs from a rechargeable battery.

[2 marks]

0 5 . 4 Explain what is meant by the 'hydrogen economy'.

[1 mark]

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0 6 . 1 Explain how ionising radiation damages living cells.

[3 marks]

0 6 . 2 Distinguish between acute and chronic effects of radiation exposure.

[2 marks]

0 6 . 3 Distinguish between **exposure** and **contamination**.

[2 marks]

0 6 . 4 A worker receives an absorbed dose of 0.002 Gy from alpha radiation.
Calculate the effective dose in sieverts.

[2 marks]

Answer _____ Sv

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0 6 . 5 State what is measured by the becquerel.

[1 mark]

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0 7 Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

Table 1

Treatment	Mean dry mass / g	Standard deviation / g
Organic manure	14.2	3.9
Inorganic NPK fertiliser	16.1	4.1

0 7 . 1 Explain what the standard deviation values tell you about the data.

[2 marks]

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0 7 . 2 Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

[2 marks]

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0 7 . 3 The test gave a probability p = 0.30. State what this means and what the student should conclude.

[2 marks]

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0 7 . 4 Explain why a statistically significant correlation does not prove a causal relationship.

[2 marks]

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0 7 . 5 Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

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0 8 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 8 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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0 9 Table 1 shows the volume of water in some reservoirs of the hydrological cycle and the annual transfer rates through them.

Table 1

Reservoir	Volume / x 10^{15} dm^3	Annual transfer (inflow = outflow) / x 10^{15} dm^3 yr^{-1}	Percentage of global water / %
Oceans	1 340 000	425	97
Ice caps and glaciers	24 000	3	1.7
Groundwater	23 000	12	1.7
Lakes and rivers	180	40	0.013
Soil moisture	17	80	0.001
Atmosphere	13	480	0.001

0 9 . 1 The residence time of water in a reservoir is given by: residence time = volume in reservoir / annual transfer rate. Use Table 1 to calculate the residence time of water in the atmosphere. Give your answer to the nearest whole day.

[3 marks]

Answer _____ days

0 9 . 2 Calculate the residence time of water in the oceans.

[1 mark]

Answer _____ years

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0 9 . 3 Explain why the atmosphere is an important reservoir even though it holds only 0.001% of global water.

[2 marks]

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0 9 . 4 Explain why human changes to a small reservoir such as soil moisture are more obvious than changes to the oceans.

[2 marks]

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0 9 . 5 Explain why the hydrological cycle is described as being in dynamic equilibrium.

[2 marks]

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1 0 Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m^{-2} day^{-1}	Mean solar energy in December / kWh m^{-2} day^{-1}
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

1 0 . 1 Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

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Answer [ ] times

1 0 . 2 Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

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1 0 . 3 Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

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1 0 . 4 Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

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1 0 . 5 Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

OR

1 1 . 2 Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS